SIMPLE MODULES OVER A FINITE ALGEBRA

THE STACKS PROJECT AUTHORS

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Editorial note. The proof is the author's complete argument for all four assertions; each is a short but genuine reduction with its reason given, and no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Finite algebras and their modules. Same chapter, the definition of a finite algebra and of a module over it, with the ground field acting. Supplies that modules are vector spaces over the ground field and that a cyclic submodule of a finite algebra is finite dimensional.
- Simple modules and their endomorphisms. Same chapter, the definitions of a simple module and of the endomorphism ring of a module. Supplies the notions in the statement and the fact that a nonzero endomorphism of a simple module has zero kernel and hence is an isomorphism.

Lemma 0.1. *Let A be a k-algebra. If A is finite, then*

(1) *A has a simple module,*
(2) *any nonzero module contains a simple submodule,*
(3) *a simple module over A has finite dimension over k, and*
(4) *if M is a simple A-module, then $End_A(M)$ is a skew field.*

Proof. Of course (1) follows from (2) since A is a nonzero A-module. For (2), any submodule of minimal (finite) dimension as a k-vector space will be simple. There exists a finite dimensional one because a cyclic submodule is one. If M is simple, then $mA \subset M$ is a sub-module, hence we see (3). Any nonzero element of $\mathrm{End}_A(M)$ is an isomorphism, hence (4) holds. □

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